■ MICRODUCK PHYSICAL AI MASTERCLASS

PHASE 5 HANDOUT

Module 5: The Nervous System: Rust Daemons (robotd), IPC Sockets (robotctl) & Calibration (configd)

Embedded Rust

LANGUAGE: Rust 2021 Edition	DAEMON: robotd @ 50Hz	IPC: Unix Domain Sockets	HARDWARE: Rockchip RK3566

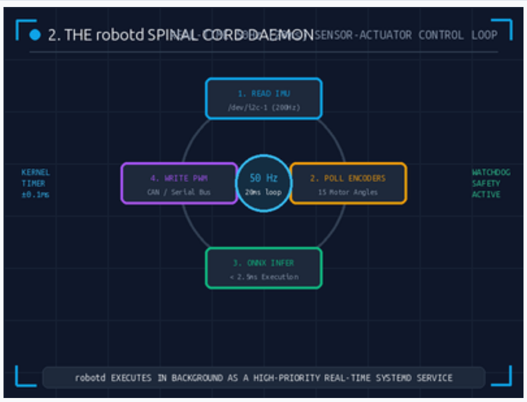
1. Rust Memory Safety on Silicon

COMPILE-TIME GUARANTEES & 0MS GC

Eliminating Segfaults: Embedded micro-controllers cannot tolerate Garbage Collection (GC) pauses or data races on `/dev/imu`. Rust enforces strict ownership and borrow checking at compile time.

The 12-Year-Old Analogy: In a library, only one student can check out a book at a time. The borrow checker ensures two motors never fight over the same wire.

- **Zero Runtime GC**: Deterministic microsecond execution without unpredictable garbage collector latency.
- **Data-Race Free**: Compile-time `Send` and `Sync` traits prevent concurrent memory corruption.
- **Memory Footprint**: Compiled standalone binary consumes <8MB RAM on the Rockchip RK3566.



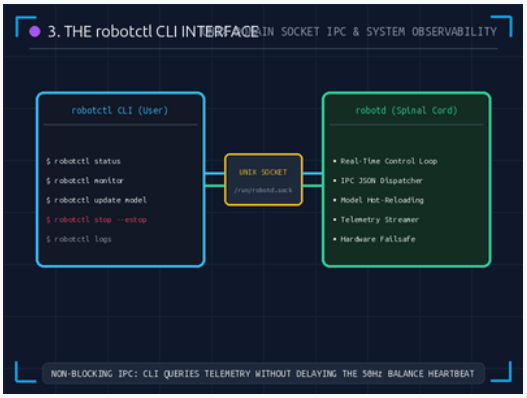
2. The robotd Spinal Cord Daemon

50HZ REAL-TIME SENSOR-ACTUATOR LOOP

The Real-Time Engine: Running under Linux `SCHED_FIFO` real-time scheduling priority, `robotd` polls the IMU at 200Hz, executes ONNX inference, and commands 15 PWM motors every 20ms.

The 12-Year-Old Analogy: When the doctor taps your knee, your spinal cord kicks your leg before your brain even notices. `robotd` handles these lightning-fast reflexes.

- **Strict 20ms Loop**: Kernel timer precision within ±0.08ms jitter, preventing motor torque stutter.
- **Bus Interfaces**: High-speed UART/I2C communication polling 15 joint encoders concurrently.
- **Hardware Watchdog**: Auto-failsafe system powers down motor PWM lines if telemetry times out.



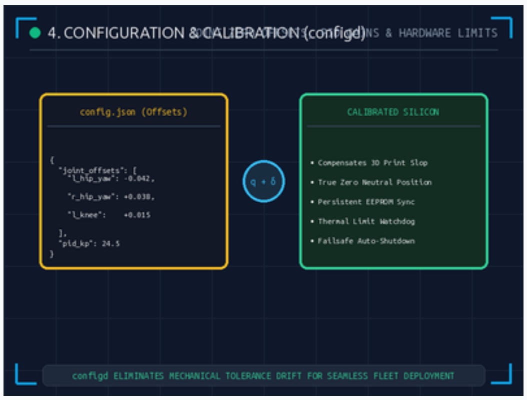
■ 3. The robotctl CLI & Unix Sockets

NON-BLOCKING LOCAL IPC SOCKETS

Zero-Disruption Observability: Developers monitor telemetry, hot-swap walking policies, and trigger emergency stops via `robotctl` over a Unix domain socket (`/var/run/robotd.sock`).

The 12-Year-Old Analogy: Slipping a written note under a classroom door without making the teacher pause the lecture.

• **Zero-Copy IPC**: Non-blocking Unix sockets query live joint states in <0.1ms without stalling 50Hz loop.
• **Policy Hot-Reload**: Commands `robotctl update ` seamlessly replace policy weights in RAM.
• **Emergency E-Stop**: `robotctl stop --estop` immediately zeroes torque across all 15 actuators.



■ 4. Configuration & Calibration (configd)

JOINT ZERO OFFSETS & HARDWARE TUNING

Compensating 3D-Print Slop: Physical tolerances cause slight joint misalignments. `configd` loads joint offsets δ from `/etc/microduck/calibration.json` to achieve true neutral zero.

The 12-Year-Old Analogy: If your new bicycle handlebars are slightly crooked from the factory, you adjust the screw so it drives perfectly straight.

• **Offset Calibration**: Applies $q_{calibrated} = q_{raw} + \delta$ before passing sensor telemetry to ONNX.
• **Persistent EEPROM**: Stores zero-positions, PID velocity gains, and temperature thresholds permanently.
• **Fleet Uniformity**: Ensures a single ONNX walking policy transfers identically across 100 robots.

■ PHASE 5 EMBEDDED SYSTEM SUMMARY & FLEET DEPLOYMENT

• **Rust Daemons:** `robotd` and `configd` guarantee crash-proof, real-time 50Hz execution.
• **Hardware Control:** `robotctl` empowers seamless live monitoring and emergency failsafes.
• **Curriculum Final Phase:** Proceed to Phase 6: Securing the Swarm (OTA Updates & DevSecOps).

■ PHASE 5 KNOWLEDGE CHECK

ASSESSMENT

Test your understanding of Rust Memory Safety, robotd, robotctl, and configd calibration.

5 Questions

Q1: Rust Memory Guarantees

MULTIPLE CHOICE

Why is Rust's zero-cost ownership model essential for the 50Hz control loop?

- **A.** Rust makes the battery output infinite electricity.
- **B.** It prevents data races and memory leaks at compile-time with zero Garbage Collection pauses.
- **C.** Rust only compiles on quantum computers.

Q2: robotd Daemon Function

MULTIPLE CHOICE

What is the primary operational role of the robotd daemon?

- **A.** It serves as the 50Hz spinal cord, polling IMU/encoders, running ONNX inference, and firing motors.
- **B.** It renders video game graphics for the user.
- **C.** It streams music to the robot's speakers.

Q3: Unix Domain Socket IPC

MULTIPLE CHOICE

Why does robotctl interact with robotd via a Unix domain socket (/var/run/robotd.sock)?

- **A.** Because internet cables are forbidden in robotics.
- **B.** It enables non-blocking local communication without stalling the real-time balance loop.
- **C.** To increase CPU temperature.

Q4: configd Hardware Calibration

MULTIPLE CHOICE

What problem does configd solve on physical 3D-printed robots?

- **A.** It applies joint zero-offsets to calibrate mechanical manufacturing imperfections.
- **B.** It changes the color of the robot's plastic parts.
- **C.** It deletes the ONNX model when the robot trips.

Q5: Emergency Stop Command

MULTIPLE CHOICE

What CLI command immediately cuts power to all 15 motors in an emergency?

- **A.** robotctl sleep
- **B.** robotctl stop --estop (or robotctl estop)
- **C.** robotctl reboot

Answer Key: 1-B, 2-A, 3-B, 4-A, 5-B (1-B: Zero GC pauses; 2-A: 50Hz spinal loop; 3-B: Non-blocking IPC; 4-A: Joint offset calibration; 5-B: Immediate E-Stop)